

Ishrak Latif

+61-489-154-254 | ishrak10319@gmail.com | **Portfolio:** ishraklatif.github.io | github.com/ishraklatif

Intro

AI Engineer with 7+ years across applied ML, enterprise software, and senior leadership — now completing an MSc in Artificial Intelligence at Monash University (full-time, graduating 2026). Specialises in production NLP, RAG systems, and ML pipeline deployment at scale. Brings consulting experience advising health ministry stakeholders and a military background managing high-stakes operations — uniquely combining technical rigour with real world leadership and client impact..

Technical Expertise

- LLM fine-tuning, RAG pipelines & vector search-LangChain, LangGraph
- ML system design & production deployment (Azure, Kubernetes)
- NLP: BERT, HuggingFace, transformer architectures
- Stakeholder advisory & technical roadmap delivery
- AI risk, responsible AI, and GDPR/data governance
- Python, PyTorch, LightGBM, FastAPI, PySpark, SQL, matplotlib

Professional Experience

Senior AI/ML Engineer

Aztech Valley Ltd

01/09/2022 – 30/07/2024

Bangladesh

- Architected and deployed a production NLP triage platform serving 200+ clinical staff, cutting average emergency response time by 20% — a direct patient-safety outcome.
- Built RAG pipelines using BioMistral and PubMedBERT with ChromaDB vector indexing, improving medical query accuracy by 25% and reducing retrieval latency by 40%.
- Engineered PySpark EHR preprocessing pipelines on Azure supporting 1.5M records/hour with distributed fault-tolerance, enabling reliable ML inference at enterprise scale.
- Automated CI/CD across Kubernetes clusters (Docker, Jenkins, MLflow), cutting manual deployment effort by 60% and accelerating model iteration cycles.
- Implemented GDPR-compliant data encryption using Azure Key Vault; achieved 100% audit pass rate across all security evaluations.

AI Consultant

Dhaka National Medical Institute Hospital & BD Thai Ltd

01/07/2023 – 30/07/2024

Bangladesh

- Advised hospital directors and government stakeholders on AI-based medical technology acquisitions, reducing procurement cycle time by 15%.
- Designed and presented AI modernisation roadmaps to the Bangladesh Health Ministry; secured ministerial approval and funding allocation for a national hospital digitisation initiative.
- Led cross-vendor integration teams during hardware deployment, coordinating technical and non-technical stakeholders with zero operational downtime.

Lieutenant (Cybersecurity & Operations)

Bangladesh Army

01/01/2016 – 25/06/2021

Bangladesh

- Led 120+ personnel in high-pressure operational environments, developing the command presence and structured decision-making relevant to leading client engagements.
- Conducted penetration testing and cybersecurity assessments on MLRS fire-control networks; identified and patched critical encrypted-channel vulnerabilities under operational conditions.
- Commended for precision in operational planning – consistently delivered outcomes on time in ambiguous, resource-constrained environments.

Public Projects

FINANCIAL MARKET FORECASTING – ASX EQUITY RETURN FORECASTING | Python, LightGBM, PyTorch

- Built a cross-sectional return forecasting system across the ASX 50 universe using a 260-feature panel (245,000+ samples, 2005–2026). Designed and evaluated 5 architectures; selected LightGBM after rigorously diagnosing deep learning generalisation failures across a structural market regime change.
- Final model: Test IC = +0.054, ICIR = +0.314, annualised L/S Sharpe = +0.429 – outperforming all neural baselines on the held-out 2023–2026 test window.
- Resolved T4 GPU OOM failure in TFT's Variable Selection Network (260 GRNs × 126 timesteps) by redesigning to a VSN-free linear encoder, reducing GPU memory ~6× while preserving the temporal attention stack.

RAG PIPELINE | Zephyr-7B, sentence-BERT, ChromaDB, LangChain

- Built a Retrieval-Augmented Generation (RAG) system integrating **Zephyr-7B** and **sentence-BERT** embeddings to provide contextual document summarization.
- The retriever achieved a **Recall@5 of 0.82**, indicating it successfully retrieved relevant biomedical passages in 82% of test queries.
- Processed **over 100K research abstracts from PubMed** for biomedical question answering and semantic search.

QUESTION CLASSIFIER — BERT PREFIX TUNING & TRANSFORMER NLP | Python, PyTorch, HuggingFace, FastAPI, React

- Built a full-stack NLP classifier (6 TREC categories) comparing a custom Transformer encoder against BERT prefix tuning. Prefix tuning achieved 94%+ accuracy while training fewer than 1% of total BERT parameters — demonstrating parameter-efficient fine-tuning at minimal compute cost. Served via containerised FastAPI + React.

Education

Monash University Expected 2026	Melbourne, Aus
Master of Science in Artificial Intelligence(Full time)	
Bangladesh University of Professionals Jan 2022	Dhaka, Bd
Bachelor of Science in Computer Science and Engineering	
Bangladesh Military Academy Dec 2018	Chittagong, Bd
Military Science	